Juan Calderon

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Education

University of California, Berkeley: Electrical Engineering and Computer Science & Mathematics August 2024 - May 2028

Relevant Coursework:

• Efficient Algorithms • Data Structures • Discrete Mathematics • Circuits & Devices • Probability & Random Processes
• PDEs and Real Analysis • Signals & Systems • Optimization Models • Computer Architecture • Linear Algebra

Skills

Languages/Systems: Python • Rust • Java • C • C++ • Assembly (RISC-V) • React • Linux
DevOps: S3 • Lambdas • EC2 • Bedrock • Azure (App Insights, Load Testing) • OpenTelemetry • K8s/Docker • CI/CD • LangGraph/LangChain • Unit & Integration Testing • Vector Databases • APIs • PostgreSQL • Node.js
Soft Skills: Team Organization • Active Listener • Pressure Resilience • Results-Driven • Eager to Learn

Experience

Microsoft, Software Engineer Intern - Redmond, WA May 2026 – August 2026

- Designed and shipped **centralized telemetry** across **6 internal AI services** with **OpenTelemetry** and **Azure Application Insights**, standardizing **P50/P90/P95 latency**, reliability, and usage metrics with strict privacy boundaries
- Built automated **cross-service correctness gates** replaying multi-step **Model Context Protocol (MCP)** workflows against **source-of-truth REST APIs** with **semantic JSON canonicalization** to detect row-level discrepancies
- Hardened a **production MCP server**: migrated to **stable MCP 1.3**, converted transport to **stateless operation** (no load-balancer affinity), and eliminated tool-call **race conditions** via single-flight recovery and session re-handshaking
- Built **LLM evaluation infrastructure**: expanded golden-set corpus with **70 verified questions**, triaged **~500 failed queries**, and shipped **QoS dashboards** with nightly pass-rate baselines and regression detection used by senior Microsoft leadership

Amazon Web Services, SDE Intern on SageMaker Team - Palo Alto, CA May 2025 – August 2025

- Built **CI/CD** pipelines deploying **ECS Fargate** services across AWS (S3, Glue, DataZone, Athena), cutting deployment time and enabling **zero-downtime** rollout.
- **Developed and evaluated** RAG agents analyzing S3 buckets with **100,000+** entries on the **new SageMaker Unified Studio**, automating client workflows; designed and deployed the client-facing portal
- Coordinated team of 6 interns, maintaining versioning and consistent documentation for a full-stack **multi-agent application**

Emory University, Researcher - Atlanta, GA June 2022 – August 2024

- Automated molecular docking pipeline with Python scripts orchestrating concurrent AutoDock Vina instances via batch job scheduling, replacing manual in-vitro screening with binding-affinity prediction across 6,000+ compounds
- Scaled pipeline to 2,000,000+ molecules by parallelizing across Emory HPC clusters
- Published findings in **Q2 journal**; co-invented a patented compound (US Patent 20240415923 A1)

Projects

Homely - Home Services Marketplace | TypeScript, Node.js, React, PostgreSQL, REST API

- Designed a dual-sided marketplace on React 19 + Express with **JWT** auth, **role-based access control**, and real-time messaging; shipped as a cross-platform PWA with **CI/CD** via **GitHub Actions & Railway**
- Engineered a **RAG**-powered AI advisor (**LangChain, Claude**) and an **MCP server** driving agentic workflows for contractor matching, job discovery, and scheduling

NBA Live-Game Trading Bot on Polymarket | Rust, tokio, WebSockets

- Built a live sports-market trading bot in **Rust** on a multi-threaded **tokio** runtime: WebSocket order-book ingestion with **lock-free snapshot sharing**, a **Poisson-MLE** signal engine with likelihood-ratio gating, and **EIP-712**-signed FOK order submission to Polymarket's CLOB
- Designed **crash-safe idempotent execution** persisting client-order IDs to **redb** before submission for restart reconciliation, plus a **kill-switch risk controller** (position caps, daily-loss limits, feed-gap detection)

News-Based, RAG-assisted Trading System | Python, C++, Qdrant, FinBERT

- Built an autonomous pipeline ingesting financial news via RSS, scoring sentiment with FinBERT, embedding into Qdrant, and generating RAG-based strategy proposals gated behind dual-window backtesting (**252d in-sample + 90d OOS**), lookahead-bias guards, and human-in-the-loop approval
- Compiled hot-path execution (backtest engine, cost model, broker, risk checks) to C++ via pybind11/CMake, validated by **24 parity tests** against Python reference implementations

Awards and Honors

- Jane Street FOCUS program participant (1 of ~40)
- Gates Scholarship Recipient (1 of 750 chosen)
- Microsoft HOLA Scholarship Recipient (1 of 13 chosen)
- H-Index of 1 (4 Citations)
- Amazon Future Engineer Scholarship (1 of 400)
- Top 5% on 2024 AMC 12B (114/150)

Publications

- Xu, E., Park, S., **Calderon, J. (2023).** In Silico Identification and In Vitro Validation of Repurposed Compounds Targeting the RSV Polymerase. Microorganisms, 11(6), 1608.
- US Patent 20240415923 A1 "RNA-Dependent RNA Polymerase Inhibitory Compounds" (Emory University, Dec 19, 2024).